

³⁶Ar(d,t) 1970Wh04,2015Fr01

$J^\pi=0^+$ for ³⁶Ar ground state.

1970Wh04: A 21.0-MeV deuteron beam was produced by the Yale MP tandem Van de Graaff accelerator. The target was a ³⁶Ar gas cell. Tritons were detected using a 140- μ m-530- μ m thick ΔE -E telescope of silicon surface barrier detectors with FWHM=65-70 keV. Measured $\sigma(E_t, \theta)$. Deduced levels, L-transfers, and spectroscopic factors from JULIE-DWBA analysis of the measured $\sigma(\theta)$. Comparisons with shell-model calculations.

2015Fr01: A 22-MeV deuteron beam was produced by the MP tandem Van de Graaff accelerator at the Maier-Leibnitz Laboratorium (MLL) in Garching, Germany. Targets were fabricated by implanting 25-70-keV 3-6 μ g/cm² of ³⁶Ar ions into 30 μ g/cm² natural abundance carbon foils. Reaction products were momentum analyzed by a Q3D magnetic spectrograph. Tritons were detected using a multiwire gas-filled proportional counter backed by a scintillator at the focal plane. Measured E_t at $\theta_{lab}=15^\circ$, 20° , 25° with FWHM $\approx$ 9 keV and at 54° with FWHM $\approx$ 16 keV. Deduced levels, proton resonance energies, level densities. Comparisons with shell-model calculations.

³⁵Ar Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where $N=3.33$ is a normalization factor adopted by 1970Wh04 from 1966Ba54.

E(level) [†]	J^π [‡]	L [#]	C ² S [#]	Comments
0	3/2 ⁺	2	3.4	
1180 10	1/2 ⁺	0	1.4	
1700	(5/2 ⁺)	(2)	<0.2	
2635 20	(3/2 ⁺)	(2)	0.5	C ² S: 1970Wh04 authors state that there is a large uncertainty in the spectroscopic strength. 1970Wh04 also gives S=0.11 or 0.032 assuming L=1.
2986 20	5/2 ⁺	2	2.6	
3200 20	7/2 ⁻	(3)	0.33,0.11	C ² S: assuming $r_{0n}=1.25$ F and $V_n\approx 60$ MeV, respectively. 1970Wh04 authors state that there is a large uncertainty in the spectroscopic strength.
5913 5				
5991 3				
6037 3				May be a doublet (2015Fr01).
6055? 3				Tentative (2015Fr01).
6076 3				
6164 3				
6253 3				
6273 3				
6302 3				
6332 3				
6345 3				
6415 2				
6439? 4				Tentative (2015Fr01).
6460 3				
6523 3				
6557 3				
6585 3				
6606 3				
6617 2				
6644 3				
6651 3				
6672 3				

[†] From 1970Wh04 for low-lying states up to 3200 keV and from 2015Fr01 for others.

[‡] As given in 1970Wh04, used for extracting C²S.

[#] From DWBA analysis of the measured $\sigma(\theta)$ in 1970Wh04.